

HE2003 Econometrics I: Linear Regression with Multiple Regressors 1

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Sep 23, 2024

Linear Regression with Multiple Regressors

By **focusing only on the student-teacher ratio**, the analysis in the simple regression ignored some potentially important factors (**the model is too simple**) by **putting them into the regression error term**.

For instance, an **omitted variable** which is particularly relevant in California because of its large immigrant population: **The fraction of the students who are still learning English (EL_Pct)** in each district.

- Students who are still learning English may perform worse on tests than native speakers (**When EL_pct goes up, Test Score goes down**).
- If districts with large class size also have many students still learning English, e.g., states with many immigrants, (**STR correlated with EL_pct**), **then the OLS regression with only the STR could be problematic**.
- In the California data, the **(sample) correlation between the STR and the percentage of English learners** in the district is 0.19 (Not a huge correlation, but already large enough to become a concern).

Linear Regression with Multiple Regressors

KEY CONCEPT: Omitted Variable Bias (OVB)

The OVB is the bias in the OLS estimator that may arise when the explanatory variable is correlated with a variable that is omitted from the analysis. The OVB occurs if the following **two conditions** hold:

1. **The explanatory variable (regressor) is correlated with the omitted variable.**
2. **The omitted variable is a determinant of the dependent variable.**

To see **the effect of the omitted variable**, we consider the following example. Suppose that the true model is

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + e_i, i = 1, \dots, n,$$

where X_{1i} and X_{2i} are **correlated**, and e_i is the **error term**. Also suppose that the coefficient **β_1 is the parameter of interest**.

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Is the OVB a serious problem?

Now, suppose that, instead of estimating the full model, we estimate the following simple regression model, and omit X_{2i} from the regression (putting it into the error term):

$$Y_i = \beta_0 + \beta_1 X_{1i} + u_i, i = 1, \dots, n,$$

where $u_i = \beta_2 X_{2i} + e_i$. That is, the error term in this regression is the sum of $\beta_2 X_{2i}$ and the true error term e_i .

The consequence is serious: The OLS estimator becomes **biased** because the OLS Assumption 1 will not hold. That is, if there is OVB, then

$$E[u_i | X_{1i}] \neq 0,$$

which implies that

$$E[\hat{\beta}_1] \neq \beta_1.$$

That is, the distribution of the OLS estimator will **no longer be centered at the true value.**

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Do the conditions for OVB hold in the following examples?

Example 1: Percentage of English Learners

Because the **percentage of English learners is correlated with the STR**, so the **first condition for OVB holds**. In addition, it is plausible that students who are still learning English will do worse on the tests than native speakers (because of the **proficiency in English**), so that the percentage of English learners is a **determinant of test scores** (negative effect on test score); Thus, the **second condition of OVB also holds**.

Example 2: Time of Day of the Test

The time of the day of the test (9am, 3pm, etc.) typically **would not be correlated** with the STR, so the **first condition does not hold**. On the other hand, it could be a **determinant of test scores** (alertness varies through the day), so the **second condition holds**.

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Comments on the OVB

- Omitted variable bias is a problem whether the sample size is large or small. The bias term **persists even in large samples**.
- Whether the bias is large or small in practice depends on the **relationship between the (included) regressor (X_{1i}) and the error term ($u_i = \beta_2 X_{2i} + e_i$)**.
- In fact, we have the following **important formula for the OVB**:

$$OVB = \frac{Cov[X_{1i}, u_i]}{Var[X_{1i}]} = \beta_2 \frac{Cov[X_{1i}, X_{2i}]}{Var[X_{1i}]}.$$

Therefore, the **sign of the OVB depends on two things**: (1) The sign of β_2 , and (2) the sign of $Cov[X_{1i}, X_{2i}]$. We can use this OVB formula to predict if the estimation result (for example, the result for STR) has **upward/downward bias**.

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Two ways to overcome omitted variable bias:

1. Run a randomized controlled experiment in which **class size (STR)** is randomly assigned among the students. Then, *EL_Pct* is still a determinant of *TestScore*, but *EL_Pct* will be uncorrelated with **class size** because of the random assignment. (This solution is **rarely feasible in economics**. e.g., parents will complain...)

Note: **How does the California data differ from a randomized experiment?** The class size is not randomly assigned, therefore **student background** can be correlated with class size (*EL_Pct* correlated with *STR*).

2. Use a **regression in which the omitted variable (*EL_Pct*) is no longer omitted** → **Multiple Regression**.

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KEY CONCEPT: The Multiple Regression Model

The multiple regression model extends the simple regression model by including **additional variables as regressors**. This model permits estimating the effect on Y_i of **changing one variable** (X_{1i}) while **holding the other regressors** (X_{2i}, X_{3i} , and so forth) **fixed** (That is to say, **controlling the other regressors**). The model can be written as

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + u_i, \quad i = 1, \dots, n.$$

For the OLS estimator to behave well, we need to impose the following **assumptions**:

1. The error term has **conditional mean zero** given $X_{1i}, \dots, X_{ki}$; that is, $E[u_i | X_{1i}, \dots, X_{ki}] = 0$;
2. $(X_{1i}, \dots, X_{ki}, Y_i), i = 1, \dots, n$ are **independent and identically distributed**;
3. **Large outliers** of data $X_{1i}, \dots, X_{ki}$ and Y_i are **unlikely**;
4. There is **no perfect multicollinearity**.

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Comments on the OLS assumptions for multiple regression

- Assumptions 1-3 are similar to those of simple regression (with one regressor).
- **The meaning of Assumption 1:** Given $X_{1i}, \dots, X_{ki}$, the average value of the **omitted factors** (u_i) is zero, and this implies that u_i is not correlated with $X_{1i}, \dots, X_{ki}$.
- **Assumption 4 is new.** The regressors are said to be **perfectly multicollinear** (or to exhibit perfect multicollinearity) if one of the regressors is a **perfect linear function of the other regressors**. For example,

$$X_{1i} + X_{2i} = 1, \text{ or } X_{1i} = 2 \times X_{2i},$$

for $i = 1, \dots, n$. If it is the case, then we will have **problem** with the OLS regression.

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Interpretation of coefficients in multiple regression

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + u_i,$$

$i = 1, \dots, n$. Consider changing X_1 by ΔX_1 while holding X_2 constant (holding X_2 unchanged).

- Population regression line **before the change**:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

- Population regression line **after the change**:

$$Y + \Delta Y = \beta_0 + \beta_1 (X_1 + \Delta X_1) + \beta_2 X_2$$

- Take the difference:**

$$\Delta Y = \beta_1 \Delta X_1$$

Therefore, $\beta_1 = \Delta Y / \Delta X_1$, holding X_2 fixed. Similarly, $\beta_2 = \Delta Y / \Delta X_2$, holding X_1 fixed.

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KEY CONCEPT: The OLS Estimator in Multiple Regression

Similar to the case with only one regressor, the OLS estimators $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$ are obtained by **minimizing**

$$\sum_{i=1}^n [Y_i - (\beta_0 + \beta_1 X_{1i} + \dots + \beta_k X_{ki})]^2$$

with respect to $\beta_0, \beta_1, \dots, \beta_k$.

The **terminology** of OLS in the multiple regression is the same as in the simple regression:

- The **OLS regression line** is the straight line constructed using the OLS estimators: $\hat{\beta}_0 + \hat{\beta}_1 X_1 + \dots + \hat{\beta}_k X_k$,
- The **predicted value** of Y_i is $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1i} + \dots + \hat{\beta}_k X_{ki}$,
- The **residual** is the difference between Y_i and its predicted value: $\hat{u}_i = Y_i - \hat{Y}_i, i = 1, \dots, n$.

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Properties of the OLS Estimators in Multiple Regression

- Under Assumptions 1-4, the OLS estimators $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$ are **unbiased**:

$$E[\hat{\beta}_j] = \beta_j, \quad j = 0, 1, \dots, k.$$

- Also, **when the sample size is large**, $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$ are **normally distributed**:

$$\hat{\beta}_j \sim N\left(\beta_j, \sigma_{\hat{\beta}_j}^2\right), \quad j = 0, 1, \dots, k.$$

- However, it is difficult to prove the results without using **matrix algebra**, so we **omit the detail**.
- The important point is that in terms of these properties, there is **nothing conceptually different** between the single- or multiple-regressor cases.

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Application to Test Scores

In the simple regression, we used OLS to estimate the regression relating test scores (**Test Score**) to the student-teacher ratio (**STR**), using the 420 observations from California school districts; the estimated OLS regression line is:

$$\widehat{TestScore} = 698.9 - 2.28 \times STR.$$

Now we include **percentage of the students in the district who are English learners** (**EL_Pct**) to the OLS regression, and we obtain:

$$\widehat{TestScore} = 686.0 - 1.10 \times STR - 0.65 \times EL_Pct.$$

Q: **What happened to the coefficient of STR?** What is the sign of the OVB? Is it **upward or downward bias?**

(Note: $Corr[STR, EL_Pct]=0.19$)

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STATA implementation with multiple regression

```
. reg testscr str el_pct, robust
```

Linear regression

Number of obs = 420

F(2, 417) = 223.82

Prob > F = 0.0000

R-squared = 0.4264

Root MSE = 14.464

testscr	Robust					
	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
str	-1.101296	.4328472	-2.54	0.011	-1.95213	-.2504616
el_pct	-.6497768	.0310318	-20.94	0.000	-.710775	-.5887786
_cons	686.0322	8.728224	78.60	0.000	668.8754	703.189